



Lighthouse prompt

A compact request with large hidden state

Design a low-power **64-bit RISC-V-based compute subsystem** for an **XR Bench-class real-time mobile XR workload**, realize it as a **vector-capable CPU, accelerator, or SoC block**, under a **3 W TDP target in a 3 nm-class LP mobile process**, and return a **design-space report with evidence and rejected alternatives**.



Architecture obligations

What the prompt immediately forces an architect to define

XR Bench workload

multi-model streams, deadlines

CPU / accelerator / SoC block

partitioning, vector support, interfaces

3 W power envelope

activity, thermal, energy/QoE tradeoff

3 nm-class physical constraints

process, timing, leakage, signoff

ISA and ABI contract

RISC-V baseline, extensions, compatibility

Memory and data movement

locality, hierarchy, interconnect energy

Compiler/runtime stack

codegen, libraries, deployment

Reliability and verification

correctness, corner cases, rejection



AI-assisted architecture study

Architecture capabilities guide represented execution and analysis

Architecture work

formulate, explore, implement, evaluate, explain, defend

Study state

Methods + tools

Evidence

Decision support

architect judgment redirects the study



Boundary condition

AI supports the architecture study through explicit state, tools, evidence, and decision support, not prompt-to-chip automation.